Chapter 10 Warm-Up: List and Dictionary Comprehensions

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1 A List of Student Names

Suppose we are given a list of student names:

```
names = ["Ana","Ben","Chloe","Diego","Elena"]
```

Student IDs begin with the eight-digit number `02601001` and are assigned sequentially. Thus, the first student receives ID `02601001`, the second receives `02601002`, etc.

2 Generating IDs with a Loop

One way to create the student IDs is

```
st_ids = []
first = 02601001
for i in range(len(names)):
    st_ids.append(first + i)
```

1. What values does `i` take?
2. Predict the contents of `student_ids`.
3. Why is `len(names)` useful here?

3 List Comprehension

Python can express the same operation more compactly:

```
st_ids = [first+i for i in range(len(names))]
```

This is called a **list comprehension**, with the pattern

```
[expression for item in sequence]
```

For each value of `i`, Python evaluates `first+i` and places the result into the list.

4 Building a Student Dictionary

Now suppose we want a dictionary in which
- the **student ID** is the key, and
- the **student name** is the value.

Using an ordinary loop:

```
students = {}
for i, name in enumerate(names):
    students[first+i] = name
```

The dictionary will contain entries such as

```
{
 2601001: "Ana",
 2601002: "Ben",
 2601003: "Chloe",
 ...
}
```

The function `enumerate(names)` provides both the index and the name for each student.

5 Dictionary Comprehension

The entire dictionary can instead be created with

```
students = {
    first+i: name
    for i, name in enumerate(names)
}
```

This is a **dictionary comprehension**, with the pattern

```
{
 key_expression: value_expression
 for item in sequence
}
```

6 Compare the Two

A list comprehension:

```
[first_id + i
 for i in range(len(names))]
```

creates a sequence of student IDs. A dictionary comprehension creates the ID-name pairs:

```
{first_id + i: name
 for i, name in enumerate(names)}
```

7 Check Your Understanding

1. Which comprehension creates only the student IDs?
2. Which comprehension associates each ID with a name?
3. Where do the names in the dictionary come from?
4. Which part of the dictionary comprehension generates the ID?
5. What would the ID for `"Diego"` be?
6. If a sixth name is added to `names`, does either comprehension need to be rewritten? Explain.